

Equal perturbation norms do not characterize maximal monotonicity: Conjecture 00000008852

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The clause being disproved. The English statement says that the perturbed operator remains maximal monotone *exactly when* its perturbation norm δ does not exceed the strong monotonicity constant γ of the original operator. We refute this necessary-and-sufficient claim for individual perturbations, even for bounded linear operators on the real Hilbert space $\mathbb{R}$, using the optimal strong monotonicity constant. The counterexamples can have arbitrarily small absolute perturbation norm. The Chinese version describes a threshold at $\delta \leq \gamma$. A *uniform worst-case guarantee* over all perturbations of bounded norm is a different statement: our example does not refute that sufficient guarantee or a sharp worst-case threshold. It refutes the pointwise “exactly when” assertion in the supplied English statement. Since this is a conjunct of the conjecture, the other trajectory assertions need not be decided.

Definitions. On $H = \mathbb{R}$, the inner product is $\langle x, y \rangle = xy$ and the norm is $|x|$. An operator F is γ -strongly monotone when

$$(x - y)(F(x) - F(y)) \geq \gamma|x - y|^2 \quad (x, y \in \mathbb{R}).$$

A relation $R \subset \mathbb{R}^2$ is monotone if $(x - y)(u - v) \geq 0$ whenever $(x, u), (y, v) \in R$. An operator is maximal monotone if its graph is monotone and has no proper monotone relational extension. Thus maximality here is not merely maximality among bounded linear operators.

Lemma 1. *For every $t > 0$, the operator $F_t(x) = tx$ is maximal monotone.*

Proof. Graph monotonicity is $t(x - y)^2 \geq 0$. Suppose a monotone relation R contains the graph, and $(x, u) \in R$. Put $y = (u + tx)/(2t)$. The point (y, ty) belongs to the graph, hence to R , so

$$0 \leq (x - y)(u - ty) = -\frac{(u - tx)^2}{4t} \leq 0.$$

It follows that $u = tx$. Every point of R already belongs to the graph, which proves full relational maximality. □

Theorem 1. *For any $a > 0$, let $A = aI$ and $B_+ = 2aI$, $B_- = -2aI$. Then:*

1. *The optimal strong monotonicity constant of A is $\gamma = a$.*
2. *Both perturbations have operator norm $\delta = 2a > \gamma$.*
3. *$A + B_+ = 3aI$ is maximal monotone.*
4. *$A + B_- = -aI$ is not even monotone.*

Proof. For A , the strong monotonicity inequality holds with equality at $\gamma = a$. Conversely, testing that inequality at $x = 1, y = 0$ forces every admissible constant to be at most a . The bounded linear scalar operator cI has norm $|c|$: its action has $|cx| = |c||x|$ for all x , and the upper bound is attained at $x = 1$. This gives both stated perturbation norms. The preceding lemma with $t = 3a$ proves the positive case. In the negative case, graph points $(1, -a)$ and $(0, 0)$ have monotonicity pairing $-a < 0$. $\square$

Arbitrarily small perturbations and zeros. Given any tolerance $\varepsilon > 0$, take $a = \varepsilon/4$. Then $\|B_+\| = \varepsilon/2 < \varepsilon$, but $\|B_+\| > \gamma = \varepsilon/4$, while $A + B_+$ is maximal monotone. This prevents an unspecified requirement of small *absolute* norm from saving the necessity claim. The unperturbed and both perturbed operators have the unique zero 0, so the example lies within the setting of operators with a zero described by the source. In particular, zero existence causes no difficulty. The opposite outcomes of B_+ and B_- show concretely why perturbation size alone cannot decide the property for a given operator.

Formalization. The Lean project uses Mathlib’s actual continuous linear maps $\mathbb{R} \rightarrow_L \mathbb{R}$ and their operator norm. It defines monotone relations and maximality by quantification over all relations on the real Hilbert space, and proves the maximality lemma directly. The theorem `same_norm_opposite_behavior` establishes the optimal constant, both exact operator norms, and the opposite monotonicity outcomes. The theorem `arbitrarily_small_counterexample` proves the tolerance statement for every $\varepsilon > 0$, and `conjectured_iff_false` negates the claimed criterion even when the constant is required to be optimal. No graph maximality or spectral fact is assumed as a certificate. The final theorem concerns the disputed conjunct, not a replacement statement about convergence or continuity of zeros.